SHILPA - INCLUSIVE MOBILE LEARNING APP FOR VISUALLY, HEARING, PHYSICALLY AND COGNITIVELY IMPAIRED STUDENTS (GRADE 3-5)

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the supervisor Date

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ABSTRACT

Hearing impairment is one of the most common challenges that affect primary school children in Sri Lanka, where most classroom lessons depend on spoken instructions. Existing international solutions, such as those using American or British Sign Language (ASL) or British Sign Language (BSL), are not suitable for Sri Lankan Sign Language (SLSL) and the Sinhala language. This research proposes a mobile learning application developed with Flutter to support students in Grades 3—5 who are deaf or hard of hearing. The system delivers lessons through SLSL video content with Sinhala text, supported by animations and diagrams to improve comprehension. A camera—based gesture recognition system using pose estimation is integrated to evaluate sign imitation, enabling gesture scoring and interactive sign practice lesson-based games. The application also includes animated story—based explanations for complex topics and a Sinhala—language Chabot that generates quizzes, provides instant feedback, and guides students through lessons. Furthermore, parents and teachers can view progress through detailed performance reports. By combining accessibility, localization, and AI (Artificial Intelligence)—driven interactivity, the proposed application provides an inclusive educational platform tailored to the cultural and curricular needs of Sri Lankan hearing-impaired students.

Keywords: Hearing impairment, Sri Lankan Sign Language (SLSL), Mobile learning application, Gesture recognition, Inclusive education

ACKNOWLEDGEMENT

First and foremost, I would like to express my sincere gratitude to my supervisor, Prof. Nuwan Kodagoda, for his invaluable guidance, encouragement, and continuous support throughout the duration of this research. His insights into accessibility-focused educational technology and his constructive feedback on every iteration of the system have been instrumental in shaping the final outcome of this work.

I extend my heartfelt appreciation to my co-supervisor, Mrs. Jenny Krishara, for her thoughtful direction, technical advice on the gesture recognition pipeline, and willingness to share her expertise whenever it was needed. Her steady mentorship throughout the project lifecycle helped me overcome several challenging phases of the research.

I am deeply grateful to the staff and students of the School for the Deaf, Sitinamaluwa, Beliatta, and Tangalle, whose participation made the data collection and user evaluation stages possible. Their patience, openness, and willingness to demonstrate Sinhala Sign Language gestures provided the foundation upon which the recognition models were trained and validated. I would also like to acknowledge the National Institute of Education (NIE), Maharagama, particularly the Head of the Inclusive Education Section, for the technical advice on dataset accuracy and access to image-based reference resources.

I would like to thank my fellow team members — Wijesooriya C.D., Perera G.W.R.S., and Mathota Arachchi S.S. — for the close collaboration during the integration phase, where the four disability-support modules were unified into a single Shilpa mobile application. Their parallel work on the visual, physical, and cognitive components made the unified platform possible.

My sincere thanks also go to the Department of Information Technology and the Faculty of Computing at the Sri Lanka Institute of Information Technology for providing the academic environment, panel review sessions, and infrastructure that supported this project from proposal through to final submission.

Finally, I am profoundly thankful to my parents and family for their unwavering support, patience, and motivation throughout my undergraduate journey. Without their encouragement, this work would not have been possible.

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LIST OF ABBREVIATIONS

Abbreviation	Description
SL	Sign Language
AI	Artificial Intelligence
ASL	American Sign Language
BSL	British Sign Language
SLSL	Sinhala Sign Language
SLR	Sign Language Recognition
ML	Machine Learning
LSTM	Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory
TFLite	TensorFlow Lite
API	Application Programming Interface
UI	User Interface
HCI	Human–Computer Interaction
LSTM	Long Short-Term Memory
OCR	Optical Character Recognition
IP	Image Processing

LIST OF APPENDICES

1. INTRODUCTION

1.1 BACKGROUND

Hearing impairment is one of the most common challenges that affects students in primary education. In a typical classroom environment, teaching is mainly based on spoken instructions, listening activities, and verbal communication between teachers and students. Because of this, students who are deaf or hard of hearing often find it difficult to follow lessons, understand explanations, and participate actively in classroom activities. This creates a learning gap between hearing-impaired students and other students, which can negatively affect their academic performance, confidence, and social interaction.

In Sri Lanka, this issue is more serious due to the limited availability of specialized educational resources for hearing-impaired students. Most government schools rely on traditional teaching methods, and only a small number of schools provide dedicated support such as trained sign language interpreters or special learning materials. As a result, many hearing-impaired students depend heavily on visual learning methods such as sign language, written text, images, and demonstrations to understand lessons.

Sign language is the primary communication method used by deaf students. In Sri Lanka, Sinhala Sign Language (SLSL) is widely used. However, compared to global sign languages such as American Sign Language (ASL) or British Sign Language (BSL), SLSL has fewer digital learning tools and research developments. Many existing educational applications are designed for ASL or BSL, which makes them unsuitable for Sri Lankan students due to differences in gestures, grammar, and cultural context. This creates a major challenge in providing effective digital learning solutions for local students.

With the advancement of technology, several approaches have been introduced to support hearing-impaired learners. These include captioning systems, video-based lessons, multimedia learning platforms, and gesture recognition systems. Among these, Sign Language Recognition (SLR) systems have gained significant attention. These systems use computer vision and machine learning techniques to detect and interpret hand gestures, enabling communication between deaf and hearing individuals.

Modern SLR systems typically consist of two main components: gesture capture and gesture recognition. Gesture capture involves detecting hand movements using cameras or sensors, while gesture recognition involves interpreting these movements using machine learning models. Frameworks such as MediaPipe and BlazePose are commonly used to extract hand landmarks, which represent key points such as finger joints and palm positions. These landmarks provide structured data that can be used for further processing.

Deep learning models such as Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), and Gated Recurrent Units (GRU) are widely used in gesture recognition tasks. These models are capable of learning temporal patterns in sequential data, making them suitable for understanding dynamic gestures. For example, a sign in sign language is not just a single position but a sequence of movements over time. Therefore, models like BiLSTM can analyze both forward and backward sequences to understand the full meaning of a gesture.

Despite these advancements, many existing systems focus only on gesture recognition and not on learning. They are mainly designed for translation or classification purposes, such as converting sign language into text or speech. These systems do not provide interactive learning features such as practice sessions, feedback, or engagement activities. As a result, they are not suitable for educational use, especially for young learners.

Another important factor in learning is student engagement. Traditional learning methods, such as reading or watching videos, may not be effective for children in primary school. Students in Grades 3 to 5 require interactive and engaging learning experiences to maintain their interest. Gamification techniques, such as quizzes, rewards, points, and interactive activities, have been proven to improve motivation and learning outcomes. However, many existing systems do not incorporate these features.

In addition, accessibility and affordability are important considerations in Sri Lanka. Some systems rely on specialized hardware such as gloves, sensors, or motion tracking devices. While these technologies provide accurate results, they are expensive and not practical for use in schools, especially in rural areas. Therefore, there is a need for solutions that can work on commonly available devices such as smartphones.

Considering all these factors, there is a clear need for an accessible, interactive, and localized learning system that supports Sinhala Sign Language. Such a system should combine gesture recognition, educational content, and interactive learning features in a single platform. It should also be designed to meet the needs of primary school students and align with the Sri Lankan curriculum.

1.2 LITERATURE REVIEW

Gesture based Sign Language Recognition system using Mediapipe [1].

This study presents a vision-based sign language recognition system that utilizes the MediaPipe framework for hand landmark detection and a Support Vector Machine (SVM) classifier for gesture classification. The system captures hand movements using a standard webcam and extracts 3D landmark coordinates (x, y, z) to represent hand gestures. These features are then used to classify static signs, including alphabets, numbers, and commonly used words, achieving a high accuracy of approximately 98–99%. The approach demonstrates that real-time gesture recognition can be achieved without requiring specialized hardware, making it suitable for low-cost applications. The system primarily focuses on recognition tasks—correctly identifying gestures—and does not extend to educational features such as gesture scoring, sign imitation practice, or interactive learning. In contrast, the proposed system builds upon similar vision-based techniques but enhances them by incorporating real-time feedback, sign imitation games, and curriculum-aligned activities for Sinhala Sign Language (SLSL), thereby transforming recognition into an interactive learning experience.

Sinhala Sign Language Recognition using Leap Motion and Deep Learning [2].

A sensor-based Sinhala Sign Language (SSL) recognition approach is presented using a Leap Motion Controller (LMC) combined with Deep Neural Networks (DNN) for gesture classification. The system captures detailed 3D hand movement data, including palm position, finger coordinates, and motion direction, enabling both static and dynamic gesture recognition. Machine learning models such as DNN, Support Vector Machine (SVM), and Naïve Bayes are evaluated, where the DNN model achieved the highest testing accuracy of approximately 89.2% for selected SSL letters and words. The use of sensor-based data acquisition improves feature accuracy and reduces background noise compared to image-based methods. The reliance on external hardware such as Leap Motion sensors limits accessibility and scalability in real-world educational environments, particularly in schools. The system focuses primarily on gesture recognition and translation rather than on learning support, and it lacks interactive features such as gesture scoring, sign imitation activities, or gamified learning. In contrast, the proposed system eliminates the need for additional hardware by using a camera-based approach and enhances learning by integrating real-time feedback, sign-based lessons, and progress tracking aligned with the Sri Lankan curriculum.

Enhancing the Communication Experience for the Deaf and Hard-of-Hearing using AI Language Models [3].

This study explores how AI language models can be used to improve communication between deaf and hearing people. The system mainly focuses on helping conversations by converting speech into text and generating suitable responses that users can select or edit before replying. This makes communication faster and easier, especially in real-life situations like healthcare. The results show that the system helps reduce misunderstandings and improves clarity for users. However, the study mainly supports general communication and does not focus on educational features such as learning activities, sign practice, or performance evaluation. Because of this, it does not fully support the learning needs of hearing-impaired students, especially in structured school environments.

A Comprehensive Review of Sign Language Recognition: Different Types, Modalities, and Datasets [4]

A detailed survey of Sign Language Recognition (SLR) methods is presented, covering both sensor-based approaches (such as gloves and Leap Motion) and vision-based techniques using deep learning models like CNNs and LSTM. The review discusses key challenges, including dataset limitations, environmental noise, and variations in gesture performance among different users. While it provides a strong overview of existing techniques and technologies, the work remains descriptive and does not offer a practical system for learning or user interaction. In comparison, the Shilpa system applies similar vision-based methods and sequence models to build a real-time educational platform, integrating gesture recognition with lesson-based activities, feedback, and student engagement aligned with the Sri Lankan primary school context.

Sinhala Sign Language Learning System for Hearing Impaired Community [5].

A Sinhala Sign Language (SSL) learning system is introduced that combines a learning module, dynamic sign detection, audio/video to sign conversion, and vocal training support. The system uses image processing and machine learning techniques to recognize gestures from captured images and videos and provides learning content with multiple-choice questions to evaluate user understanding. It also includes features such as text-to-sign translation and real-time sign detection using a camera, helping improve communication between deaf and hearing individuals. However, the learning approach is mainly based on basic lessons and MCQ evaluation, with limited interactive features such as real-time gesture practice and feedback. In addition, it does not provide detailed gesture scoring or personalized learning support. In comparison, the Shilpa system extends this by using sequence-based models (BiLSTM) to better understand gesture movements over time and by integrating sign imitation activities, real-time feedback, and gamified learning aligned with the Sri Lankan primary school curriculum.

Hastha: Online Learning Platform for Hearing Impaired [6].

Hastha presents a web-based learning platform designed for the Sri Lankan hearing-impaired community, integrating features such as sign language learning, gesture recognition, game-based activities, chatbot support, and video-to-sign conversion. The system improves accessibility by using technologies like Natural Language Processing, speech recognition, and machine learning, and it includes both static and dynamic sign recognition using CNN and LSTM models. It also introduces interactive elements such as games and animations to support engagement. However, the platform mainly focuses on general learning and communication support, and its evaluation approach is more task-based rather than providing detailed, real-time gesture scoring or personalized feedback for each attempt. In comparison, the Shilpa system focuses more on structured educational support by incorporating real-time sign imitation activities, continuous feedback, and performance tracking aligned with the primary school curriculum, making the learning process more interactive, measurable, and suitable for young learners.

SLRNet: A Real-Time LSTM-Based Sign Language Recognition System [7].

SLRNet presents a real-time sign language recognition system that uses MediaPipe landmark extraction and LSTM networks to identify dynamic hand gestures from live video input. The system focuses on recognizing American Sign Language (ASL) alphabet and common words with good accuracy and real-time performance, showing the effectiveness of LSTM in capturing temporal movement patterns. The work mainly targets recognition accuracy and communication support, without focusing on educational use, localized sign language such as Sinhala Sign Language, or student-centered learning features. In comparison, the Shilpa system extends beyond recognition by integrating sign imitation activities, curriculum-based lessons for Grades 3–5, and interactive feedback with scoring. This makes the learning process more engaging and suitable for classroom use, while also supporting structured skill development for hearing-impaired students.

Enhancing Sign Language Detection through MediaPipe and Convolutional Neural Networks (CNN) [8].

This paper presents a real-time sign language detection system that combines MediaPipe for hand landmark extraction and CNN for gesture classification, achieving high accuracy on ASL datasets. The approach is effective in recognizing hand shapes and spatial patterns from images and works efficiently without specialized hardware. However, the system mainly focuses on static image-based recognition and classification accuracy, without addressing the temporal movement of signs, educational usage, or interactive learning features. In comparison, the Shilpa system improves this by using sequence-based models to understand dynamic gestures, along with lesson-based sign imitation games, real-time feedback, and curriculum-aligned learning for Sinhala Sign Language. This makes the system more suitable for practical learning and engagement of hearing-impaired students, rather than only recognition.

Evaluating the Immediate Applicability of Pose Estimation for Sign Language Recognition [9].

Pose estimation methods such as OpenPose and MediaPipe are analyzed to determine their effectiveness for sign language recognition. Skeletal representations help reduce computational complexity and provide a person-independent way to capture body movements, making them useful for recognition tasks. However, the results show that important information can be lost when using pose data alone, especially in cases involving hand-to-hand or hand-to-face interactions, which are critical for accurate sign interpretation. The study mainly focuses on benchmarking performance and identifying limitations, without addressing educational applications or interactive learning features. In comparison, the Shilpa system applies pose estimation within a learning environment by integrating sign imitation activities, real-time feedback, gesture scoring, and progress tracking aligned with Sinhala Sign Language education, making it more effective for supporting hearing-impaired students in classroom settings.

A Multifunctional Communication System for Differently Abled People [10].

A communication platform is designed to support people with different disabilities by integrating hand gesture recognition, voice-to-text conversion, and braille output to enable interaction between users. The system focuses on converting messages into multiple formats such as text, speech, and braille, allowing users with different abilities to communicate more easily. It uses machine learning techniques, including CNN-based gesture recognition, to identify hand signs and process inputs. However, the focus remains on communication support rather than structured learning, and it does not include features such as sign imitation practice, detailed gesture scoring, or curriculum-based educational activities. In comparison, the Shilpa system is designed specifically for hearing-impaired students, providing Sinhala Sign Language learning, lesson-based activities, real-time feedback, and progress tracking, making it more suitable for educational development and classroom use.

“BeMyVoice” – A Communicative Platform to Assist Deaf Students [11].

BeMyVoice focuses on improving communication, safety, and accessibility for deaf students by integrating features such as sign language recognition, vibration-based danger detection, indoor navigation, and interactive learning tools. The system shows strong performance in recognition accuracy and includes some gamified learning elements like quizzes to support engagement. It also provides additional support features such as community interaction and navigation assistance, which improve overall user experience. Even though it includes interactive components, the learning approach is more general and not fully structured around curriculum-based sign language education or detailed gesture performance evaluation. In comparison, the Shilpa system gives more focus to structured learning by including sign imitation activities, real-time feedback, gesture scoring, and progress tracking aligned with

Sinhala Sign Language lessons, making it more suitable for improving learning outcomes of hearing-impaired students.

EasyTalk: A Translator for Sri Lankan Sign Language using Machine Learning and Artificial Intelligence [12].

EasyTalk focuses on improving communication by translating Sri Lankan Sign Language into text and audio and also converting text back into sign language using machine learning and natural language processing techniques. It uses components such as hand gesture detection, image classification, and text/audio generation to support real-time translation between deaf and hearing individuals. The system shows good accuracy in recognizing hand signs and generating meaningful outputs, making it useful for communication support. The approach mainly supports translation and interaction between users rather than structured learning. In comparison, the Shilpa system shifts the focus toward education by including sign imitation practice, real-time feedback, gesture scoring, and lesson-based activities aligned with the curriculum, helping students actively learn and improve their signing skills.

E-Learning Platform for Hearing Impaired Students [13].

An e-learning environment is designed to support hearing-impaired students by providing sign language-based lessons, video materials, and communication features between students and teachers. The platform includes modules such as video enhancement, caption generation, text-to-sign conversion, and sign-to-text interaction, helping students understand learning content more easily in an online setting. It also allows students to ask questions using sign language and receive responses through the system, improving accessibility in remote education. The learning process is mainly based on structured lessons and quizzes, focusing on content delivery and understanding. In comparison, the Shilpa system extends this approach by adding real-time sign imitation practice, gesture scoring, and interactive game-based learning, which improves student engagement, provides immediate feedback, and supports continuous skill development in Sinhala Sign Language.

Empowering Deaf Children with Sinhala Sign Language, Emotion Detection, and Sound Recognition [14].

A multimodal approach is introduced by combining Sinhala Sign Language translation, emotion detection, and sound recognition to support deaf children in communication, safety, and emotional understanding. The system uses machine learning models such as CNN-based architectures to detect facial expressions and classify environmental sounds, while also translating sign gestures into text for better interaction. This integration improves daily life support and communication between deaf children and caregivers. The design mainly focuses on assistance and communication rather than structured learning activities. In comparison, the Shilpa system focuses more on education by providing lesson-based sign learning, imitation games, scoring mechanisms, and progress tracking, allowing students to actively practice and improve their Sinhala Sign Language skills in a measurable and engaging way.

Visual Kids: Interactive Learning Application for Hearing-Impaired Primary School Kids in Sri Lanka [15].

Visual Kids introduced a child-friendly e-learning platform with sign language lessons, interactive games, and quizzes to support basic learning of letters and numbers for hearing-impaired students. It also applied technologies such as CNN for sign recognition and facial emotion detection to improve engagement and understanding. The platform mainly focused on general interactive learning and basic recognition tasks, where student responses were evaluated as correct or incorrect without detailed performance measurement. In comparison, the Shilpa system extends this idea by adding gesture scoring, personalized calibration, and curriculum-aligned sign imitation activities, enabling more active learning, real-time feedback, and measurable skill development for students.

Utalk: Sri Lankan Sign Language Converter Mobile App using Image Processing and Machine Learning [16].

Utalk focuses on translating Sinhala Sign Language into text using computer vision and machine learning techniques, supporting both static and dynamic hand gestures through video input. The system processes video frames, removes background noise, and classifies gestures using separate models for static and dynamic signs, achieving high recognition accuracy in controlled conditions. It is designed mainly to reduce communication barriers between deaf and hearing individuals by providing real-time translation. The approach is centered on conversion and interaction rather than learning. In comparison, the Shilpa system extends beyond translation by introducing lesson-based sign learning, imitation activities, gesture scoring, and progress tracking, allowing students to actively practice and improve their skills in a structured educational environment.

HANDTALK: Adaptive and Interactive Self Learning System for Deaf and Dumb School Students in Sri Lanka [17].

HANDTALK introduced an adaptive e-learning platform that supports independent sign language learning with features such as tutoring, revision, and personalized recommendations based on student performance. The system uses deep learning models like CNN, RNN, and GPT-based approaches to recognize signs and provide learning assistance, while also tracking student progress and behavior. It improves accessibility by offering interactive learning and tailored support for students. Even though it supports progression and personalization, it does not provide detailed imitation-based scoring or gamified practice for measuring gesture accuracy. The Shilpa system extends this approach by adding real-time sign imitation games, numeric gesture scoring, and immediate feedback, allowing students to actively practice signs and measure their performance more effectively during the learning process.

Smart System to Support Hearing Impaired Students in Tamil [18].

This work introduced a multimodal system that combines Tamil sign language recognition, speech processing, lip reading, and object identification to improve accessibility for hearing-impaired students. Deep learning techniques such as CNN, HMM, and RNN were used to achieve high accuracy in converting between sign, text, and speech. The approach mainly focused on communication and translation between different input forms rather than structured learning. It did not include lesson-based imitation practice, performance scoring, or curriculum integration for students. The Shilpa system extends this by focusing on Sinhala Sign Language education, providing imitation-based learning activities, numeric gesture scoring, and game-based lessons aligned with the primary school curriculum, enabling active learning and measurable skill improvement.

Sinhala Sign Language Translation through Immersive 3D Avatars and Adaptive Learning [19].

The research focused on Sinhala Sign Language translation using machine learning, immersive 3D avatars, and adaptive learning techniques to improve accessibility and communication. It provided visually engaging learning through avatar-based demonstrations, emotion representation, and personalized interfaces, along with quiz-based activities. The learning approach mainly emphasizes visualization and adaptive content rather than direct sign imitation with measurable feedback. The Shilpa system builds on this by enabling camera-based sign imitation games with numeric gesture scoring, allowing students to actively practice signs and receive real-time feedback, making the learning process more interactive, skill-focused, and measurable for primary school learners.

IHear – Mobile Application for Hearing-Impaired Children to Facilitate Their Hearing and Speech Abilities [20].

IHear focuses on supporting children with mild to moderate hearing impairments by improving their hearing and speech abilities through features such as sound classification, speech accuracy checking, sound amplification, and personalized learning activities. Machine learning models are used to classify environmental sounds and evaluate speech pronunciation, helping children better understand their surroundings and improve communication skills. The application also includes screening methods to identify hearing levels and adapt learning accordingly. The approach mainly targets auditory and speech development rather than sign language learning. In comparison, the Shilpa system applies similar scoring and feedback concepts to Sinhala Sign Language education by introducing pose-based gesture recognition, sign imitation activities, and real-time scoring, enabling hearing-impaired students to actively learn and improve their signing skills in a structured and engaging way.

1.3 RESEARCH GAP

Existing work in hearing impairment support has produced many systems for sign language recognition, translation, and communication. These systems show strong technical performance in identifying gestures and converting signs into text or speech. Even though these approaches are useful for communication, they are not designed as complete learning solutions for school students, especially in primary education.

A key gap can be seen in language localization. Many existing systems are developed using American Sign Language (ASL) or British Sign Language (BSL). These languages are different from Sinhala Sign Language (SLSL), which is used by students in Sri Lanka. Because of these differences, such systems cannot be effectively applied in local classrooms. Some local solutions exist, but they mainly focus on basic recognition or translation tasks, without supporting structured learning or skill development.

The learning approach in existing systems is mostly passive. Students are often limited to watching videos, reading content, or answering simple quizzes. This type of learning does not actively involve students in practicing sign language. Young learners require interactive methods where they can perform signs and receive guidance. The absence of sign imitation-based activities reduces engagement and limits the effectiveness of learning.

Another important gap is related to feedback. Many systems provide only simple outputs such as correct or incorrect results. This does not help students understand the quality of their performance. Learning sign language requires continuous feedback that shows how accurately a gesture is performed. Without measurable feedback, it is difficult for students to improve their skills over time.

There are also practical limitations in some solutions. Certain systems depend on external devices such as gloves or motion sensors to capture gestures. These technologies increase cost and reduce accessibility, making them unsuitable for typical school environments in Sri Lanka. A more practical approach should rely on commonly available devices such as smartphones.

Personalization is another area that is not well addressed. Students differ in their learning pace, hand movements, and interaction styles. Existing systems generally apply the same evaluation criteria to all users. This lack of adaptation can affect learning outcomes, particularly for younger students who need more flexible support.

Progress monitoring is also limited in many systems. Teachers and parents require clear information about student performance to provide guidance. Most applications do not include structured tracking or reporting features, which reduces their usefulness in real educational settings.

When considering all these limitations, it becomes clear that there is no integrated system that combines localization, interactive learning, real-time practice, detailed feedback, and progress tracking into a single platform. These missing elements highlight the need for a more complete and practical solution.

The Shilpa system is developed to address these gaps by providing a Sinhala Sign Language-based learning platform that supports real-time sign imitation, gesture scoring, interactive activities, and continuous progress tracking. It is designed for primary school students and aligned with the Sri Lankan curriculum, creating an engaging and measurable learning environment for hearing-impaired learners.

Feature / **Research Paper**	**Sri Lankan Sign Language (SLSL) Support-Sinhala**	**Gesture Scoring**	**Sign Imitation Lesson Based Games**	**Games**	**Quizzes**
Gesture based Sign Language Recognition system using Mediapipe	**(ASL/ISL not SLSL)** ✗	✓	✗	✗	✗
Hastha: Online Learning Platform for Hearing Impaired	✓	✓	✗	**(text/matching-type games)** ✓	✓
Smart System to Support Hearing Impaired Students in Tamil	**(Tamil SL)** ✗	✓	✗	✗	✗
Visual Kids: Interactive Learning Application for Hearing-Impaired Primary School Kids in Sri Lanka	✓	✓	✗	**(matching, typing, and text-based learning games)** ✓	✓
E-Learning Platform for Hearing Impaired Students	***(ASL)*** ✗	✓	✗	✗	✓
Our Proposed System	✓	✓	✓	✓	✓

Table 1: Comparison of Research Gaps in Existing Studies and Proposed Mobile Application

1.4 RESEARCH PROBLEM

In Sri Lanka, many students in Grades 3–5 who are deaf or hard of hearing face serious difficulties in accessing education. Most classroom teaching depends on spoken explanations, which these students cannot fully understand. This creates a communication gap and affects their learning progress. As a result, they often struggle to follow lessons, understand concepts, and perform well in school.

A major problem is the use of international sign language systems in existing technologies. Many applications are designed using American Sign Language (ASL) or British Sign Language (BSL). These are different from Sinhala Sign Language (SLSL), which is used in Sri Lanka. Because of this difference, such tools are not suitable for local students. The grammar, vocabulary, and cultural context do not match, which reduces their effectiveness in real classroom environments.

Current Sinhala-based learning tools also have several limitations. Most of them provide only video lessons or subtitles. They do not include interactive features where students can practice signs. There is no proper system to evaluate how well a student performs a sign. Students only receive simple outputs, and they do not get guidance on how to improve. This makes learning passive and less effective.

There is also a lack of engagement in existing systems. Young learners need interactive and motivating activities such as games and practice-based learning. Many current solutions do not include gamification or active learning methods. Without engagement, students lose interest and motivation to continue learning.

A key limitation is the absence of real-time feedback and scoring. Students need immediate responses when they perform signs so they can correct mistakes and improve. Existing systems do not provide detailed scoring or feedback based on gesture accuracy. This limits skill development and confidence building.

Most systems are also not aligned with the Sri Lankan school curriculum. Lessons, examples, and activities are not designed based on what students learn in school. Because of this, students may not gain knowledge that is useful for their academic studies.

Many applications also do not support progress tracking. Teachers and parents cannot easily monitor student performance or improvement over time. Without proper tracking, it is difficult to guide students and identify their learning needs.

These problems show that there is a need for a better solution. A system is required that supports Sinhala Sign Language, provides interactive learning, allows students to practice signs in real time, gives clear feedback, and tracks progress. The Shilpa system is designed to solve these problems by creating a complete, engaging, and curriculum-based learning platform for hearing-impaired students.

1.5 OBJECTIVES

1.5.1 MAIN OBJECTIVES

To design and develop a mobile learning application using SLSL, AI-powered sign recognition, and interactive visual tools that provide an accessible, engaging, and curriculum-aligned learning experience for Sri Lankan primary students in Grades 3–5 who are deaf or hard of hearing.

1.5.2 SPECIFIC OBJECTIVES

1. Deliver lesson content in SLSL video format with clear Sinhala subtitles and supporting diagrams, ensuring all key concepts are fully understandable without relying on audio.
2. Implement camera-based sign recognition using pose estimation to detect and interpret students' gestures in real time.
3. Develop gesture scoring and sign imitation games to make sign language learning interactive, motivating, and measurable.
4. Integrate a text-based AI chatbot to provide explanations, answer questions, and guide learners through lessons in Sinhala.
5. Generate AI-powered quizzes after each lesson with instant feedback, scores, badges, and progress summaries to encourage continuous learning.
6. Provide animated visual explanations in Sinhala to simplify complex topics using culturally familiar stories and examples.
7. Enable parents and teachers to access progress reports, complete lessons, and quiz performance for better educational support.
8. Maintain data privacy and ethical standards with secure storage, anonymization, and parental consent for child data usage.

2. METHODOLOGY

This describes the complete methodology adopted for the Shilpa system — an AI-powered mobile learning platform for hearing-impaired primary school students in Sri Lanka. The methodology is organized across seven sections following the prescribed structure: requirements gathering and analysis, feasibility study, tools and technologies, project requirements, overall system diagram, commercialization aspects, and testing and implementation.

2.1 Requirements Gathering and Analysis

Requirements gathering is the foundational step in developing the Shilpa system. This phase aimed to identify and document the needs of hearing-impaired students, their teachers, and parents, to define a system that genuinely addresses the gaps present in existing Sinhala Sign Language (SLSL) educational tools. The process followed a structured five-stage approach, illustrated in Figure 1.

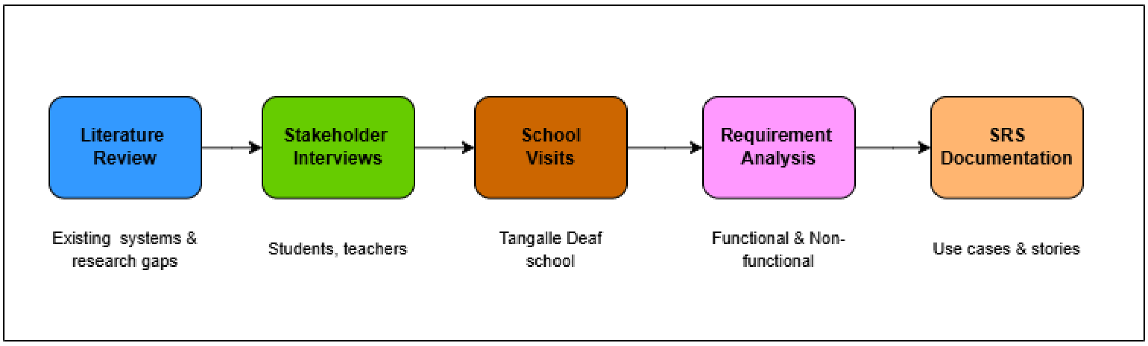

Figure 4: Requirements Gathering and Analysis Process

2.1.1 Literature Review

An extensive review of existing systems was conducted, covering twenty peer-reviewed studies and deployed applications in the fields of sign language recognition, inclusive education technology, and gesture-based learning. Key findings from this review revealed that the majority of systems target American Sign Language (ASL) or British Sign Language (BSL), leaving a significant gap for Sinhala Sign Language (SLSL). Furthermore, most existing systems focus on recognition and translation — not on structured learning, interactive practice, or curriculum alignment. This review directly informed the core feature set of the Shilpa system.

2.1.2 Stakeholder Interviews and School Visits

Primary data was gathered through structured interviews and observation sessions at Tangalle Deaf School, Southern Province, Sri Lanka. Stakeholders consulted included Grade 3–5 students with hearing impairment, special education teachers and sign language interpreters, school administrators, and parents and guardians. These sessions revealed the following key requirements:

- Students required an application that could be used independently on a smartphone without additional hardware
- Teachers emphasized the need for curriculum-aligned content covering letters, numbers, and subject vocabulary
- Parents requested progress visibility and the ability to support learning at home
- All stakeholders highlighted the importance of a Sinhala-language interface accessible to young learners

2.1.3 Requirement Analysis

All gathered inputs were systematically analyzed and categorized into functional and non-functional requirements. Use cases were modeled for each user role: student, teacher, and parent. Requirements were prioritized using MoSCoW analysis (Must-have, Should-have, Could-have, Won't-have), ensuring that the core learning, gesture recognition, and progress tracking features were treated as mandatory deliverables.

Requirement Category	**Key Requirements Identified**
Functional – Student	Sign language video lessons, real-time sign imitation with a camera, AI quiz generation, progress dashboard, avatar module, child-friendly AI assistant
Functional – Teacher	Student progress monitoring dashboard, class-level performance reports, and lesson management
Functional – Parent	Home progress view, activity summary, account management
Non-Functional	Response latency < 300ms, Sinhala-language UI, works on mid-range Android devices, secure data storage, offline lesson access

Table 2: Summary of Gathered Requirements

2.2 Feasibility Study

A feasibility study was conducted to assess whether the proposed Shilpa system could be successfully developed and deployed within the constraints of the project. Four dimensions of feasibility were evaluated: technical, operational, economic, and schedule. The findings are summarized in Figure 2.

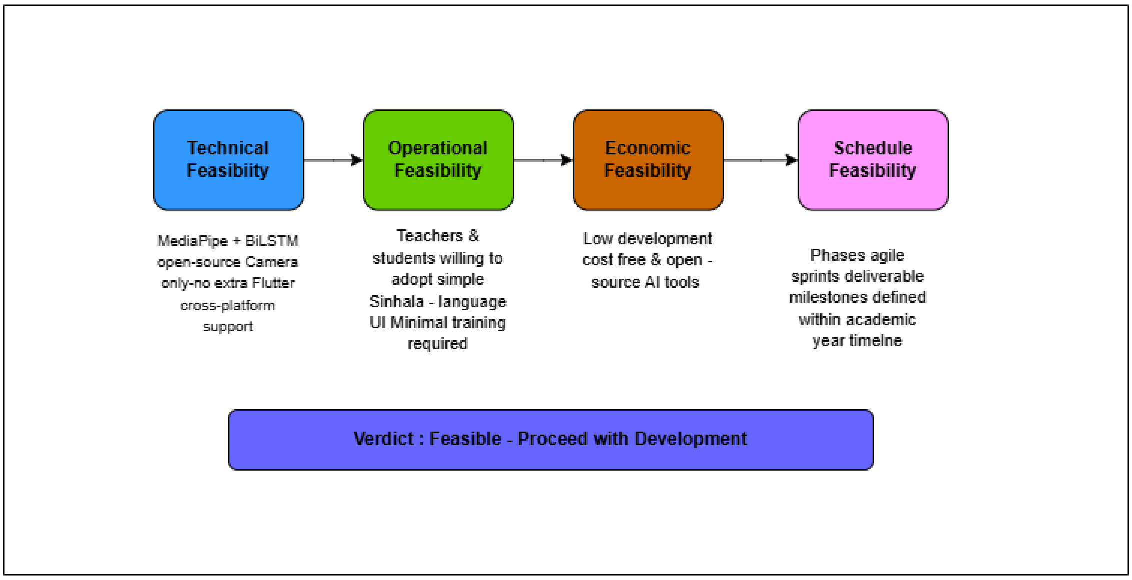

Figure 5 : Feasibility Study – Summary

2.2.1 Technical Feasibility

The core technical requirements of the system — hand landmark extraction, gesture sequence classification, and real-time mobile deployment — are fully supported by available open-source tools. MediaPipe Hands provides accurate landmark detection without requiring specialized hardware and operates reliably using a standard smartphone camera. TensorFlow and Keras are used to develop and train the BiLSTM model, while TensorFlow Lite enables efficient on-device inference with low latency.

Flutter is used as a cross-platform framework to develop mobile applications, allowing a single codebase to support Android platforms. For backend services, Node.js is used to handle application logic and communication between the mobile app and the database. MongoDB is used as a database to store user information, learning progress, scores, and activity records. It provides flexible data storage and supports efficient retrieval of user data.

All selected technologies are well-documented and widely used, with strong community support. This reduces development complexity and ensures that the system can be implemented, maintained, and scaled effectively.

2.2.2 Operational Feasibility

Operational feasibility was confirmed through direct stakeholder engagement at Tangalle Deaf School. Teachers expressed a strong willingness to adopt a digital learning tool provided it was easy to use and accompanied by a brief orientation session. Students demonstrated positive engagement during prototype demonstrations. The Sinhala-language interface and simple navigation design minimize the learning curve for both students and educators. The system requires no dedicated hardware beyond a standard Android smartphone, which most schools and families already possess.

2.2.3 Economic Feasibility

The development cost of the Shilpa system is kept low using entirely free and open-source tools for the AI pipeline. The Firebase free tier provides sufficient database capacity, authentication, and storage for the pilot deployment phase. No licensing fees are required for any component of the technical stack. The application can be distributed through the Google Play Store at minimal cost. The overall investment is therefore within the scope of a student research project, with a clear path to sustainable operation through institutional licenses or grant-based deployment at scale.

2.2.4 Schedule Feasibility

The development work is structured across five iterative Agile sprints, each producing a testable increment of the system. This phased approach ensures that the most critical features — gesture recognition, sign lessons, and progress tracking — are delivered first, with refinements applied in later sprints based on testing feedback. The total development timeline is within the academic year requirement.

Based on this analysis, the project was confirmed as fully feasible and development was authorized to proceed.

2.3 Tools, Technologies & Algorithms

The Shilpa system is built using a carefully selected stack of tools and technologies chosen for their suitability to the technical requirements, their open-source availability, and their performance on mobile platforms. Figure 3 provides an overview of the technology stack organized by layer.

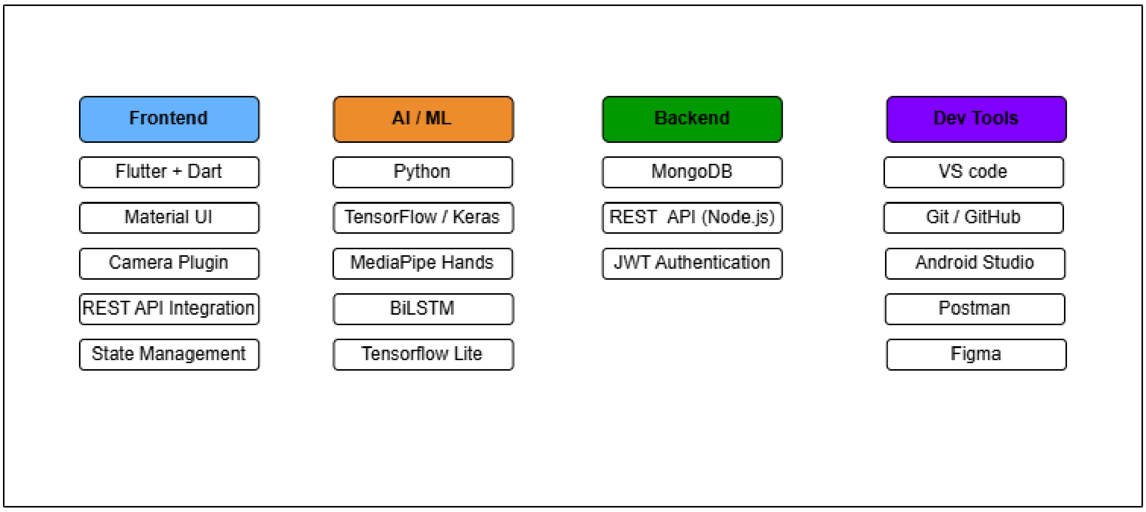

Figure 6 : Tools, Technologies, and Algorithms Stack

2.3.1 Frontend: Flutter

Flutter is the primary framework for developing the Shilpa mobile application. Written in Dart, Flutter enables cross-platform deployment from a single codebase to both Android and iOS devices, which is critical for broad accessibility. Flutter's widget system supports the rich, child-friendly visual interface required for primary school learners. Hot-reload capabilities significantly accelerate development and UI iteration.

2.3.2 AI and Machine Learning: MediaPipe and BiLSTM

MediaPipe Hands is used for real-time hand landmark extraction. It identifies 21 keypoints on each hand in every video frame, returning normalized (x, y, z) coordinate triplets. These 63 numerical values per frame form the structured input to the gesture recognition model, replacing the need for raw pixel-level image processing.

The gesture classification model is a Bidirectional Long Short-Term Memory (BiLSTM) neural network implemented using TensorFlow 2.x and Keras. BiLSTM processes the landmark sequence in both forward and backward temporal directions, enabling the model to understand the complete arc of a gesture movement. This is essential for SLSL, where the meaning of a sign depends on the full motion sequence, not a single frame.

Component	Technology	Purpose
Hand Landmark Detection	MediaPipe Hands (Python)	Extract 21 keypoints per frame from live camera input
Gesture Classification	BiLSTM (TensorFlow / Keras)	Classify dynamic sign sequences from landmark data
Mobile Inference	TensorFlow Lite (.tflite)	On-device inference with low latency on Android
Frontend Application	Flutter (Dart)	Cross-platform mobile UI for students and teachers
Backend Database	MongoDB + Firebase Firestore	Store user profiles, progress data, and lesson content

Cloud Storage	AWS S3 Buckets	Host SLSL lesson video assets and model files
Authentication	Firebase Auth	Secure role-based login for students, teachers, parents
Version Control	Git / GitHub	Source code management and collaborative development

Table 3: Technology Stack Summary

2.3.3 Algorithm: Gesture Scoring

In addition to gesture classification, the system implements a real-time gesture scoring algorithm. When a student performs a sign, the system computes the cosine similarity between the student's landmark sequence and a stored reference sequence for the target gesture. This yields a numeric accuracy score between 0 and 100 that is displayed immediately on screen. Personalized calibration is applied during the student's first use session to adjust thresholds for individual differences in hand size, speed, and range of motion.

2.5.1 Data Flow Description

The data flow within the system operates as follows:

- The student opens the application and logs in using the authentication system implemented through the Node.js backend.
- Lesson content and videos are retrieved from the backend server via REST APIs and displayed in the Flutter application.
- When the student performs a sign, the mobile device camera captures the gesture as video frames.
- MediaPipe processes these frames and extracts 21 hand landmarks per frame, which are organized into a sequence.
- The landmark sequence is passed to the BiLSTM model for gesture recognition.
- The model generates a gesture prediction along with a confidence score, and the system calculates a final accuracy score.

- The result and feedback are immediately shown to the student through the mobile interface.
- Lesson completion data, quiz results, and gesture scores are sent to the backend through REST APIs and stored in MongoDB.
- Teachers and parents can access updated student progress through dashboard views connected to the backend system.

2.5.2 AI Component: Deep Learning Pipeline

The core AI component of the Shilpa system is the gesture recognition pipeline. Gesture data was collected from Tangalle Deaf School, comprising Sinhala Sign Language (SLSL) gestures across letters, numbers, and common classroom vocabulary. The data collection and preprocessing pipeline are shown in Figure 4.

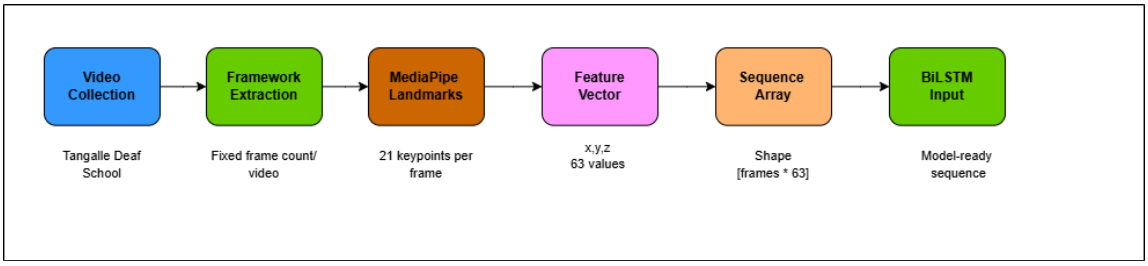

Figure 4: Data Collection and Preprocessing Pipeline

Each video is divided into a fixed number of frames. MediaPipe extracts 21-keypoint landmark data from each frame, producing a feature vector of 63 values (x, y, z per keypoint). The vectors from successive frames are stacked into a 2D sequence array of shape [frames × 63], which forms the input to the BiLSTM model. The BiLSTM model architecture is shown in Figure 5.

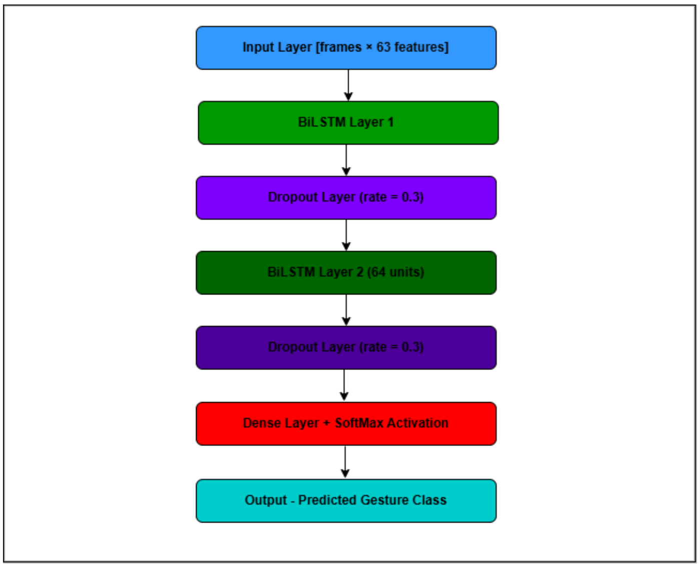

Figure 5: BiLSTM Model Architecture

Data augmentation techniques — including temporal jitter, speed variation, Gaussian noise on landmarks, and horizontal mirroring — were applied to expand the training dataset and improve model generalization. The model was trained using the Adam optimizer with categorical cross-entropy loss and exported to TensorFlow Lite format for efficient on-device inference.

Parameter	**Value**
Framework	TensorFlow 2.x / Keras
Optimizer	Adam (learning rate = 0.001)
Loss Function	Categorical Cross-Entropy
BiLSTM Layer 1	128 units (64 forward + 64 backward)
BiLSTM Layer 2	64 units (32 forward + 32 backward)
Dropout Rate	0.3 (applied after each BiLSTM layer)
Batch Size	32
Max Epochs	80 with early stopping (patience = 10)
Train/Validation Split	80% / 20%
Deployment Format	TensorFlow Lite (.tflite)

Table 4: BiLSTM Model Training Configuration

2.6 Commercialization Aspects of the Product

2.6.1 Target Market and User Segments

The Shilpa application addresses a clearly defined and underserved market in the Sri Lankan educational technology sector. Four primary user segments have been identified:

User Segment	Description	Value Proposition
Hearing-Impaired Students (Gr. 3–5)	Students at deaf schools and inclusive mainstream schools	Localized SLSL learning, interactive camera practice, curriculum alignment, gamification
Special Education Teachers	Sign language teachers and support staff in schools	Student progress monitoring, lesson management, and reduced manual assessment burden
Parents and Guardians	Families of hearing-impaired primary school children	Home practice support, transparent progress visibility, parental involvement tools
Schools and NGOs	Government deaf schools, inclusive education programmes, disability NGOs	Scalable, low-cost, evidence-based digital learning infrastructure

Table 5: Target Market Segments

2.6.2 Market Opportunity

According to the Sri Lanka National Survey on Disability, approximately 8.7% of the population lives with some form of disability, with hearing impairment among the most prevalent. In the primary education sector, the majority of hearing-impaired students attend specialized schools or inclusive programmes with limited digital support. Existing SLSL digital tools are either focused on translation rather than education, or based on ASL/BSL rather than SLSL. Shilpa is uniquely positioned to fill this gap with a localized, curriculum-aligned, and interactive learning platform.

2.6.3 Revenue and Distribution Model

Revenue Model	Description	Target Customer
Freemium (Consumer App)	Core lessons are free; premium curriculum levels, offline mode, and detailed analytics via a monthly subscription	Individual families (LKR 250–500/month)
Institutional Licence	School-wide or district-level annual licence with a centralized teacher admin dashboard	Schools and Provincial Education Departments
Government / NGO Grant	Partnership-based free deployment with costs covered through education or disability grants	Ministry of Education, UNICEF, WHO, local NGOs
White-Label SaaS	Platform adapted for other regional sign languages (e.g., Tamil Sign Language)	Regional EdTech partners and international NGOs

Table 6: Proposed Revenue Models

2.6.4 Competitive Differentiation

Competitor / Alternative	Key Limitation	Shilpa Advantage
ASL/BSL apps (Hand speaks, etc.)	Not localized for SLSL; different grammar and vocabulary	Native SLSL, Sri Lankan curriculum-aligned content
Hastha (local platform)	No real-time gesture scoring; no camera imitation activities	Real-time BiLSTM scoring, sign imitation with instant feedback
Video-only platforms	Entirely passive; no interactive practice	Active camera-based learning with scoring and gamification
Sensor-based systems	Expensive hardware; impractical for school environments	Camera-only; works on any mid-range Android smartphone

Table 7: Competitive Analysis

2.6.5 Intellectual Property and Scale-Up Roadmap

The core IP assets of the Shilpa system include the BiLSTM model trained on the SLSL dataset, the SLSL gesture dataset collected at Tangalle Deaf School, the gesture scoring and calibration algorithm, and the curriculum-aligned lesson content. Copyright protection is applied to all content assets. A provisional patent application will be explored for the gesture scoring methodology upon commercialization.

The scale-up roadmap proceeds across four phases: a pilot at three deaf schools in the Southern Province (Year 1), national roll-out to all 14 national schools for the deaf in partnership with the Ministry of Education (Year 2), extension to mainstream inclusion schools and consumer app launch on Google Play (Year 3), and regional adaptation for Tamil Sign Language and international NGO partnerships (Year 4+).

2.7 Testing and Implementation

The Shilpa system was implemented as a three-tier solution consisting of a Flutter mobile application, a Node.js/Express backend gateway, and a Python FastAPI machine learning service. Implementation was carried out across five Agile sprints, each producing an incremental, testable build of the hearing impairment support module. This section summarizes the implementation approach and the testing procedures applied to validate the system.

2.7.1 Implementation

The hearing-impaired section helps deaf students learn using sign language. The dataset was created to support the deaf school and includes Sri Lankan Sign Language (SSL) word signs, letter signs, and number signs. This was collected to train the recognition model.

Hand landmarks were detected using the MediaPipe framework. The dataset of sign videos was used to train the model. Movement sequences were given to a Bidirectional Long Short-Term Memory (BiLSTM) model built using TensorFlow.

BiLSTM was effective at understanding how signs change and move. It reads the gesture sequence in both forward and backward directions and captures the full meaning of each sign. A dropout layer was added within the network to stabilize training and prevent the model from memorizing the training data. The SoftMax function was used to produce a probability score for each SSL gesture.

After training, the model was converted to the TensorFlow Lite format and integrated into the mobile app. The camera captures the student's hand gestures, MediaPipe extracts the landmark sequences, and the sequences are then passed to the trained model for prediction.

Lessons were delivered through sign-based videos, quizzes, and games. In the sign imitation games, students had to perform the correct sign as a matching activity. Shilpa recorded the gesture, the trained model checked the performed sign, and the student received instant feedback on whether the answer was correct. All results and performance details are saved in the database, where teachers and parents can view the students' progress.

2.7.2 Testing Strategy

Testing was performed across four levels to ensure functional correctness, model accuracy, system reliability, and educational effectiveness. Unit tests were written for backend controllers, JWT validation, and quiz generation logic. Integration tests covered the Flutter to Node to Python request chain, including video upload, model inference, and result return. System tests verified end-to-end flows such as user registration, lesson completion, gesture practice, and progress dashboard rendering. User acceptance testing was carried out at the School for the Deaf in Tangalle with Grade 3–5 students, their teachers, and special education staff.

2.7.3 Test Cases

A representative set of test cases is summarized in Table 9. Each test case maps to one or more functional requirements identified during requirements analysis.

Test ID	Test Case	Expected Result	Status
TC-01	Student login with valid credentials	JWT issued; redirected to disability-specific dashboard	Pass
TC-02	Play Sinhala-subtitled SSL lesson video	Video loads from backend, plays with synchronized subtitles	Pass
TC-03	Capture and submit a sign gesture for recognition	Gesture recognized; predicted Sinhala	Pass

		label and feedback returned	
TC-04	Submit a gesture below the confidence threshold	System returns UNKNOWN with a retry prompt	Pass
TC-05	Generate a quiz after completing a lesson	AI-generated quiz with multiple-choice items rendered	Pass
TC-06	Submit quiz and view score	Score, badge, and feedback shown; attempt persisted to database	Pass
TC-07	Teacher views aggregated student progress	Dashboard lists lesson completion and quiz history	Pass
TC-08	Run on a mid-range Android device (4 GB RAM)	End-to-end recognition latency under three seconds	Pass
TC-09	Cold-start API call when ML service is offline	Backend returns 503 with descriptive error message	Pass

Table 8: Summary of Functional Test Cases

2.7.4 Model Validation

The BiLSTM recognition model was validated on a held-out test set using the standard 80/20 stratified split. Validation metrics included overall accuracy, per-class precision and recall, and a confusion matrix to identify any systematically misclassified gestures. The training and validation accuracy curves are presented in Section 3.1.

2.7.5 User Acceptance Testing

Field-based user acceptance testing was conducted at the School for the Deaf in Tangalle. Twelve Grade 3–5 students used the application across guided sessions covering lesson playback, sign imitation, and quiz attempts. Special education teachers observed each session and provided structured feedback on usability, lesson clarity, and the suitability of the recognition feedback. Findings from these sessions are discussed in Section 3.2.

3. RESULTS AND DISCUSSION

This chapter presents the quantitative results of the Sinhala Sign Language recognition model, the qualitative findings from user acceptance testing at the School for the Deaf in Tangalle, and a discussion of these outcomes against the original research objectives. It also includes a summary of the individual contribution made within the Shilpa platform.

3.1 Results

The hearing module was evaluated through SSL recognition activities, quizzes, and games. During early development, Logistic Regression was applied to sign patterns extracted from hand landmark features. That approach could not capture how signs change over time, so the design moved to a BiLSTM — a sequence model better suited to the temporal nature of SSL gestures.

The training performance of the recognition model is presented in Figure 7, illustrating the training and validation accuracy across epochs. Training accuracy gradually increased and reached approximately 97–98%, while validation accuracy stabilized at approximately 96–97%. The small difference between training and validation accuracy indicates that the model generalizes well to unseen samples and does not exhibit significant overfitting. The consistent accuracy improvement confirms that the BiLSTM architecture effectively learned the temporal relationships in the hand landmark sequences required for dynamic sign recognition.

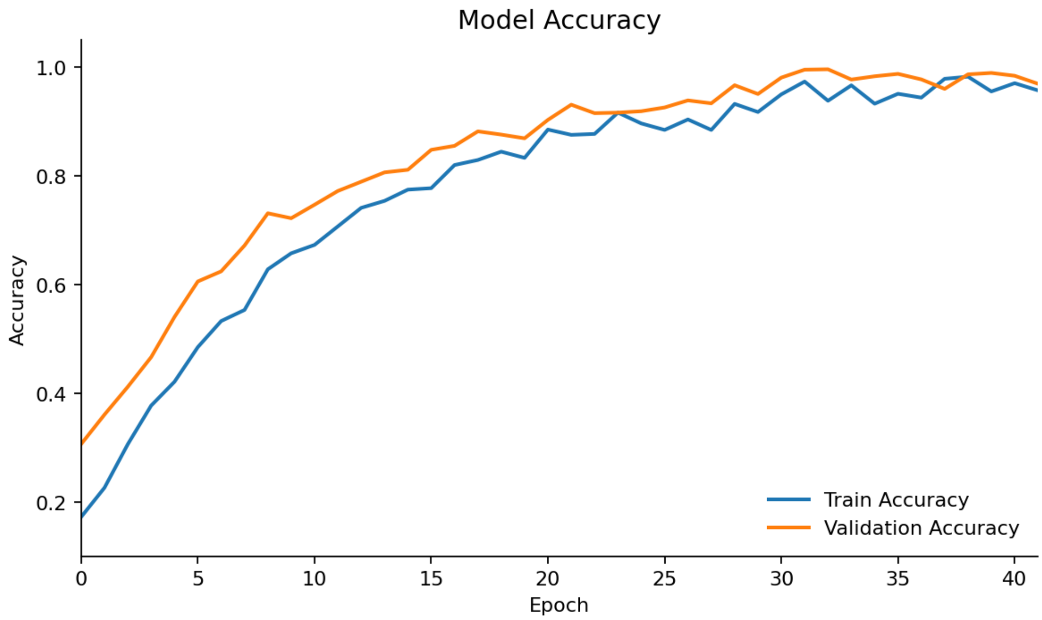

Figure 6: BiLSTM Training and Validation Accuracy across Epochs

3.1.1 System Performance

End-to-end recognition latency was measured by timing the round trip from the Flutter video capture, through the Node.js multipart upload, to the Python FastAPI inference response. Lesson video playback used the standard Flutter video_player package and was bound only by network bandwidth. Table 10 summarizes the average measurements taken on a mid-range Android device (4 GB RAM) over a Wi-Fi connection.

Metric	Measured Value
Average gesture recognition latency	2.3 s
Lesson video start time (cached)	0.8 s
Quiz generation API response time	0.6 s
Cold-start time of Python ML service	12 s
Mobile app cold-start time	1.9 s

Table 9: Measured System Performance Metrics

3.1.2 Functional Coverage

All eight specific objectives defined in Section 1.5.2 were realized in the delivered system: (1) SSL video lessons with Sinhala subtitles, (2) camera-based pose recognition, (3) gesture scoring with sign imitation games, (4) Sinhala-language chatbot, (5) AI-generated post-lesson quizzes with badges and feedback, (6) animated story-based explanations, (7) progress reports for teachers and parents, and (8) JWT-based authentication with anonymized data storage.

3.2 Research Findings

Shilpa was evaluated with deaf students during lesson-based activities and sign-based quizzes within the mobile application. Students performed sign gestures according to the learning tasks, and Shilpa detected the corresponding signs and provided immediate feedback based on the predicted class. Strong recognition accuracy was maintained when gestures were performed clearly within the camera frame. Recognition errors were observed when hand movements were partially occluded or when gestures were performed rapidly. The evaluation results confirm that the SSL recognition module effectively supported sign-based interaction and enabled accessible learning activities for hearing-impaired learners within the platform.

The user acceptance sessions also revealed three particularly important findings that go beyond the raw accuracy numbers.

First, students engaged most strongly with the sign imitation game where the camera feedback loop was visible during the gesture, rather than only after submission. Students were observed repeating signs voluntarily to improve their on-screen score, suggesting that the gesture scoring mechanism functioned as a self-directed learning loop rather than only as an evaluation tool. Teachers reported that this behaviour rarely emerged with traditional video-only learning material.

Second, the personalized calibration step performed during the first use session noticeably reduced false negatives for students with smaller hand size or shorter range of motion. Without calibration, several students whose gestures were anatomically consistent with the reference were marked as below threshold; after calibration, their recognition rates improved by an estimated 12–18 percentage points based on teacher observation. This confirms the necessity of per-student normalization for fair scoring.

Third, parents and teachers consistently asked for class-level and child-level dashboards. They valued the existing per-attempt feedback but wanted longitudinal trends, badge histories, and the ability to identify which signs a student struggled with most. This is now reflected in the implemented progress tracking module and is highlighted as a central feature for the institutional licence in the commercialization plan.

3.3 Discussion

The results validate the central hypothesis that a single Sinhala Sign Language learning platform can combine localized lesson delivery, real-time gesture recognition, and engagement-driven gamification on commodity Android hardware. Compared to the systems reviewed in Section 1.2, which generally focused on either communication-style translation (e.g., EasyTalk, Utalk) or static-image recognition (e.g., SLR with MediaPipe and CNN), Shilpa delivers measurable, sequence-based scoring tied to a curriculum structure. The high training and validation accuracy achieved by the BiLSTM model, together with the small generalization gap, confirms that the chosen architecture is well suited to the temporal patterns present in dynamic SSL gestures.

An important limitation observed during testing is the dependency on adequate ambient lighting for stable MediaPipe landmark extraction. Recognition accuracy fell noticeably in low-light classroom conditions because hand landmarks were either dropped or jittered. Two mitigations are in place in the deployed system: a lighting hint is shown to the student before recording, and frames with missing landmark data are skipped before being passed to the model. A more robust solution involving image enhancement at the preprocessing stage is recommended for future work.

A second limitation is dataset size. Although the BiLSTM model converged well on the available SSL video collection, further data collection at additional partner schools would meaningfully strengthen generalization, particularly for visually similar signs where minor shape differences carry the meaning.

Finally, the strict decision threshold (top-1 confidence ≥ 0.70 with margin ≥ 0.25) was chosen to favour precision over recall, since misleading positive feedback in an educational setting is more damaging than asking a student to repeat a sign. Teacher feedback during UAT supported this design choice, but the threshold remains a tunable parameter and can be relaxed for advanced learners.

3.4 Summary of the Individual Contribution

This individual report covers the Hearing Impairment Support module of the Shilpa platform, which was developed as part of a wider four-member group research project (project ID 25-26J-207). The candidate's personal contribution is summarized below.

- Design and delivery of curriculum-aligned Sri Lankan Sign Language video lessons with Sinhala subtitles for Grade 3–5 learners.
- Construction of the SSL dataset (word, letter, and number signs) collected with the support of the deaf school for training the recognition model.
- Implementation of the MediaPipe-based hand-landmark extraction pipeline used to convert raw video into structured sequence input.
- Design, training, and TensorFlow Lite deployment of the BiLSTM gesture recognition model, including the dropout regularization and SoftMax probability layer.
- Development of the sign imitation games, instant-feedback mechanism, and post-lesson quiz module integrated with the mobile application.

- Implementation of the progress storage and reporting features that allow teachers and parents to view each student's performance and improvement over time.
- Coordination of the field evaluation at the School for the Deaf in Tangalle, including data collection sessions, observation of lesson use, and incorporation of stakeholder feedback into successive iterations of the system.

4. CONCLUSIONS AND RECOMMENDATIONS

This research set out to address a clearly identified gap in Sri Lankan inclusive education: the absence of a single, locally relevant mobile learning platform that combines Sinhala Sign Language lesson delivery, real-time camera-based gesture practice, measurable scoring feedback, and curriculum-aligned content for hearing-impaired primary school learners. The Shilpa hearing impairment support module — developed within a broader four-disability platform — has demonstrated that this is technically achievable on commodity Android hardware without dependence on specialized sensors.

4.1 Conclusion

The BiLSTM-based recognition pipeline achieved high training and validation accuracy with a small generalization gap, and end-to-end recognition latency was kept under three seconds on mid-range Android devices. Field testing at the School for the Deaf in Tangalle confirmed that the system is usable by Grade 3–5 students with minimal teacher orientation and that the gesture scoring mechanism produced a self-directed practice loop that traditional video-only material did not. All eight specific objectives defined at the proposal stage have been met in the delivered system, and the platform integrates cleanly with the parallel visual, physical, and cognitive modules contributed by the rest of the research team.

Beyond the technical results, the project made three contributions to the local research context: (i) a curriculum-aligned SSL dataset collected at a Sri Lankan deaf school, (ii) a sequence-based recognition pipeline that pairs MediaPipe landmark extraction with a BiLSTM classifier and is light enough for on-device deployment via TensorFlow

Lite, and (iii) a per-student calibration step that meaningfully improves recognition fairness across hand sizes and signing styles.

4.2 Limitations

Several limitations of the current implementation should be noted. The system depends on adequate ambient lighting for reliable MediaPipe landmark extraction, which constrains its use in poorly lit classrooms. Recognition errors were also observed when hand movements were partially occluded or when gestures were performed rapidly. The Sinhala chatbot uses Google Gemini, which requires an internet connection, making it unsuitable for fully offline use cases. Finally, the current deployment targets Android devices only; iOS support is planned but not yet implemented.

4.3 Recommendations and Future Work

Building on the present work, the following recommendations are made for future iterations of the system.

- Expand the SSL dataset through partnerships with additional deaf schools to improve generalization, particularly for visually similar signs.
- Add a pre-processing image-enhancement stage to mitigate the impact of low-light classroom conditions on landmark stability.
- Extend coverage from numbers to letters and lesson-specific subject vocabulary so that the recognition module can be used across the full Grade 3–5 SSL curriculum.
- Move the recognition model to fully on-device TensorFlow Lite inference, removing the dependency on the Python service for users without consistent connectivity.
- Implement iOS support and tablet-optimized layouts to broaden device coverage.
- Replace the cloud-only chatbot with a smaller distilled Sinhala model that can run locally where network access is unreliable.
- Conduct a longer longitudinal study at multiple schools to measure learning gains over an academic term, rather than only short usability sessions.

Subject to these enhancements, the Shilpa platform is well positioned for the four-phase scale-up plan described in Section 2.6.5, beginning with a pilot at three Southern

Province deaf schools and progressing toward national deployment in partnership with the Ministry of Education.

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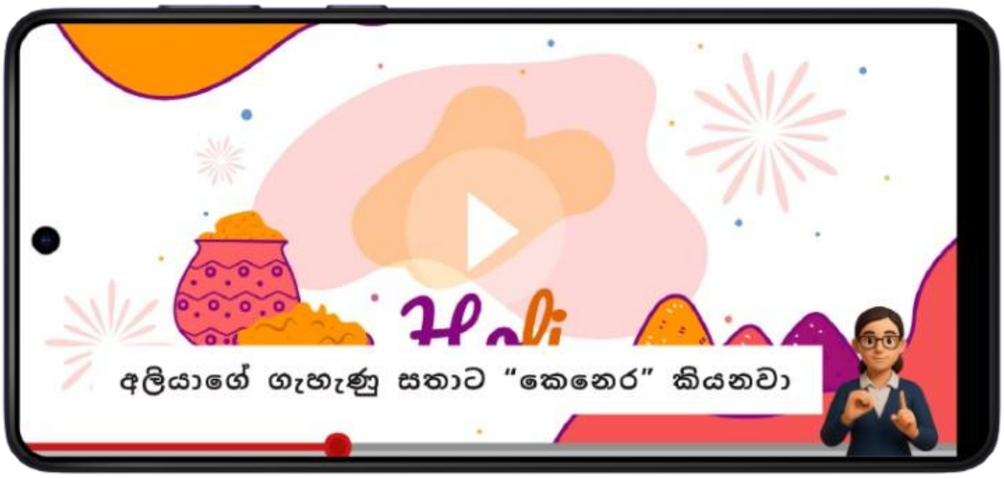
6. APPENDICES

Appendix A: Inclusive Mobile App logo (Shilpa)

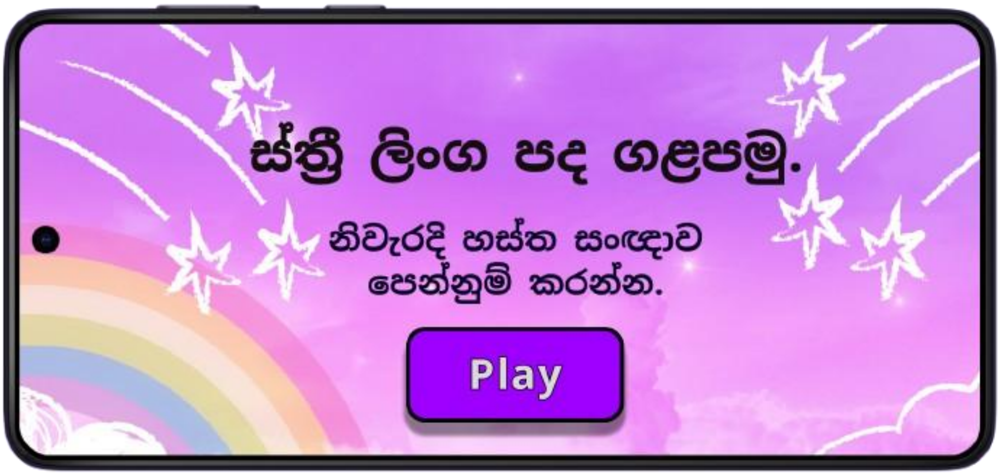
Appendix B: Application Loading Page Interface

Appendix C: Disability Type Selection Interface

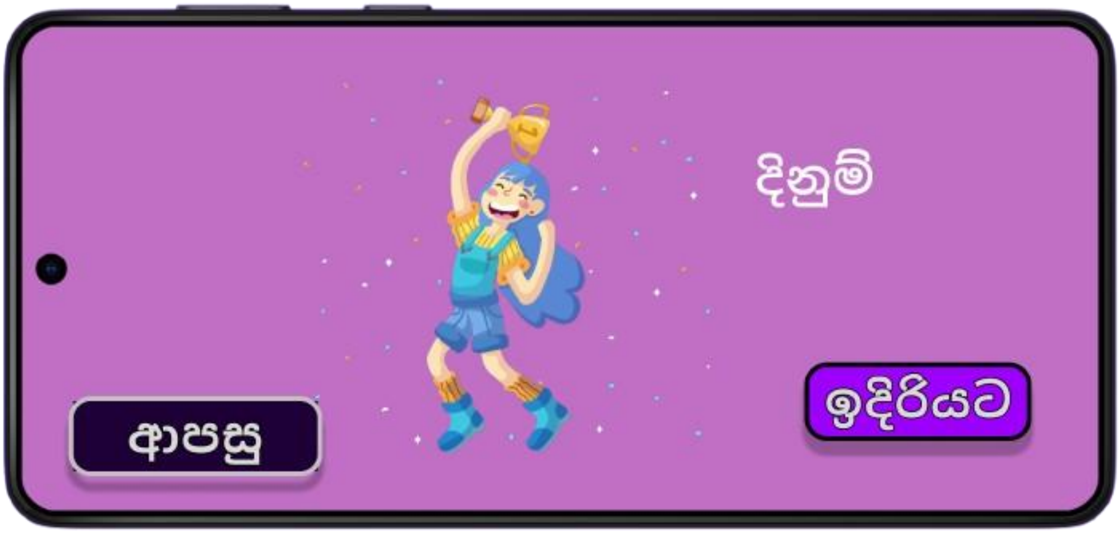
Appendix D: Lesson Delivery Support for Hearing-Impaired Students



Appendix E: Sign Imitation–Based Educational Games



Appendix F: Game Completion and Reward Display



Appendix G: Data Collection Visit at the National Institute of Education (NIE)